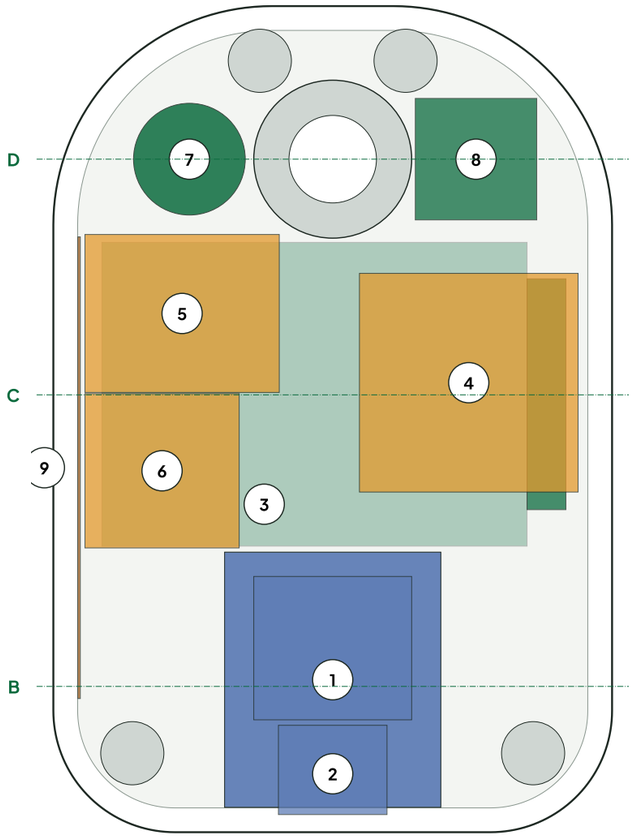


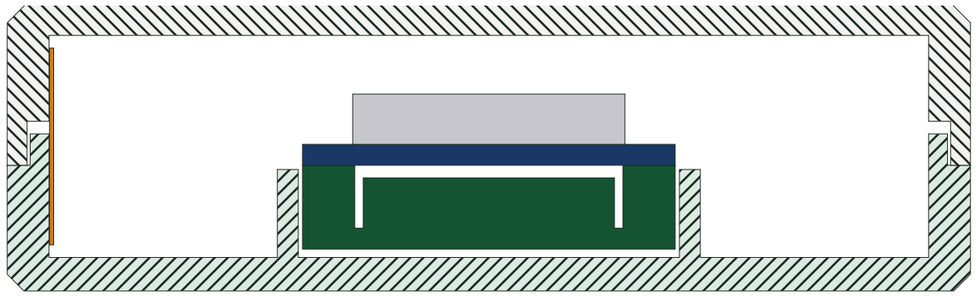
INTERNAL LAYOUT — FRONT VIEW, COVER REMOVED

scale 2.3 : 1

full depth (10 mm)
ear-side layer
front layer (behind logo)
side wall

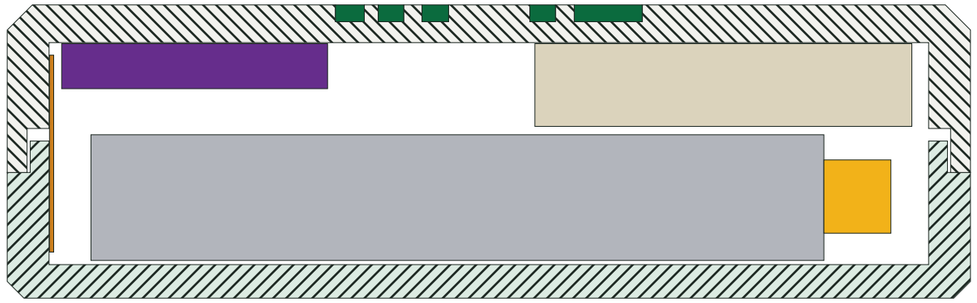
#	Component	Envelope (mm)	Function
1	Seeed XIAO ESP32S3	21.0 × 17.8 × 4.3	MCU, Li-ion charger, USB-C
2	Seeed Wio-SX1262 (XIAO kit)	21.0 × 17.8 × 4.0	LoRa / LoRaWAN AS923, u.FL
3	LiPo 602535 + PCM	35 × 25 × 6.0	3.7 V ~500 mAh, JST-SH
4	GNSS ceramic patch	18 × 18 × 4.0	passive, u.FL, faces out
5	GNSS module L76K / ATGM336H	≤ 16 × 13 × 2.4	GPS + BeiDou, UART
6	6-axis IMU breakout	≤ 12.7 × 12.7 × 2.2	MPU-6050 class, I²C
7	MLX90614ESF (TO-39)	Ø9.2 × 5.5	IR skin temp, I²C
8	MAX30102 mini module	≤ 10 × 10 × 3.2	HR + SpO_2 (PPG), I²C
9	LoRa FPC antenna 915/923	≤ 40 × 10 × 0.2	adhesive flex, u.FL

B–B



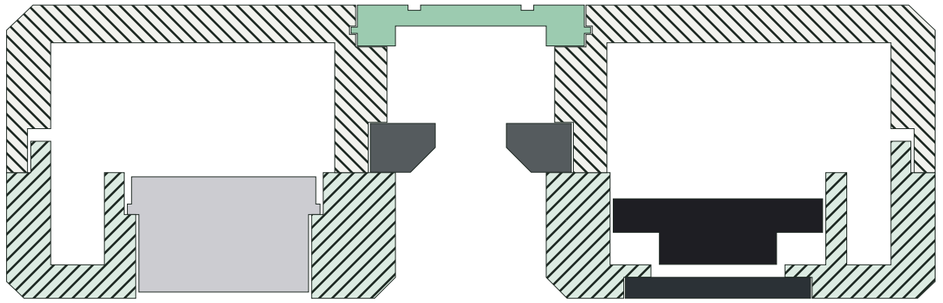
SECTION B–B · MCU / radio zone · Y = 12

C–C



SECTION C–C · battery + GNSS zone · Y = 36

D–D



SECTION D–D · pin + body-sensor zone · Y = 55.4

Terrava

SMART EAR TAG · TV-ET1

Internal layout & cross-sections

Every part checked for zero interference in CAD · Internal volume 27 cm³

UNITS	SCALE	TOLERANCE	REV	DATE	SHEET
mm	2.3 : 1 / 2.5 : 1	±0.2 unless noted	A	2026-10-21	2 / 3